Question Paper ID: 635176

Course Code: CS234AI

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RV College of Engineering

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III Semester B.E. Regular / Supplementary Examination February/March 2026

COMMON TO CD / CS / CY

Applied Digital Logic Design and Computer Organization

Time : 3 Hours **Maximum Marks : 100**

Instructions to the students

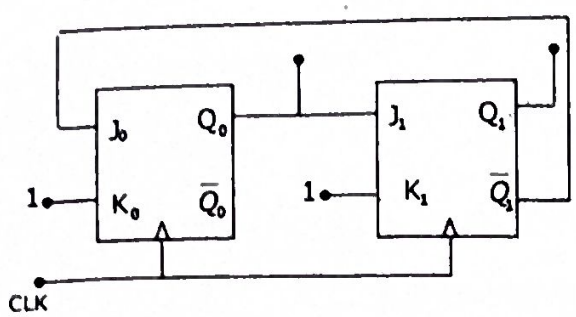
1. Answer all questions from Part A . Part A questions should be answered in first three pages of the answer book only.
2. Answer Five full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

Part A

Question No	Question	M	CO	BT
1.1	What indicates overflow in 2's complement addition?	01	1	2
1.2	Indicate the Range of a Signed 4-bit number?	01	1	2
1.3	Find out the number of prime and essential prime implicants for the given K-map	01	1	2

AB \ CD	00	01	11	10
00	0	0	X	0
01	1	0	X	X
11	X	X	1	0
10	0	X	1	0

1.4	Before the start of the Restoring Division algorithm, what initial values must be loaded into the Accumulator (A) and the Q register?	01	1	2
1.5	Draw the Logic diagram for Master-Slave Flip-Flop with Data Lockout and How can a data lock-out flip-flop be symbolically distinguished from a pulse-triggered flip-flop in digital circuit representations?	02	2	3
1.6	Derive the Characteristic Equation for JK Flip-Flop.	02	2	3
1.7	Figure shows a mod-K counter, identify the K value.	02	3	3



Draw the sequential circuit diagram for the equations given:

1.8
- $D_1 = \bar{x}\bar{Q}_2 + \bar{Q}_1 Q_2$
- $D_2 = x\bar{Q}_1 + \bar{Q}_1 Q_2$
- $z = \bar{x}Q_1 + x\bar{Q}_1\bar{Q}_2$

02 3 3

1.9 Write the meaning of the following instructions and contents of the registers R4,R2 after the execution of the following instructions/program. Assume following values before the execution of the program, R4=0X0005, R2=0X0004,[0X0004] = 0X30, [0X0005] = 0X40, [0X0009] = 0X50. 02 1 3

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ADD  R2 ,  R4 ; R2 + R4 -> R2
ADD  (R4) , R2 ;
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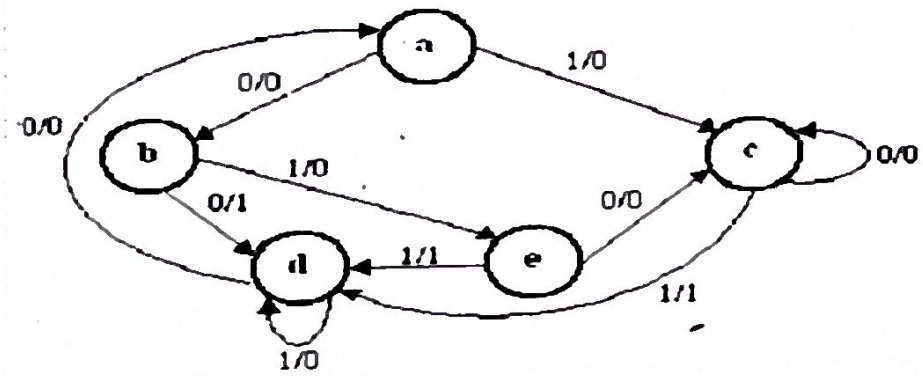
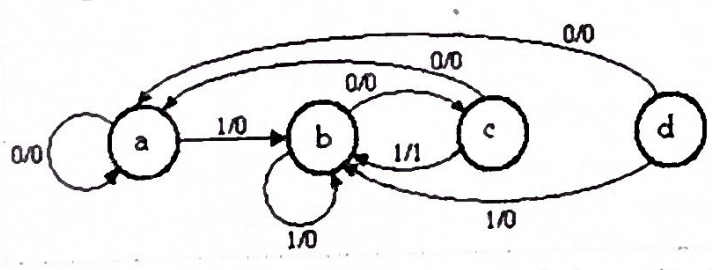
1.10 Indicate the contents of memory locations (both memory address and contents) for little endian and big endian storage formats for 32 bit number, 0x10213367, stored from memory address 3000 (assume byte addressable memory and word length 32bit) 02 4 3

1.11 Indicate the Name and Function of following Registers PC,IR,LR and SP 02 4 2

1.12 Assume a computer has 32 bit addresses. Each block stores 16 words. A direct mapped cache has 256 blocks. In which block (line) of the cache would we look for each of the following addresses. Addresses are given in hexadecimal, i) 1C2B C052 and ii) AFFB 00EF 02 4 4

Part B

Question No	Question	M	CO	BT
2a	Using a 4-variable K-map, minimize the Boolean function F(A,B,C,D)=Σm(0,1,2,5,7,8,9,10,13,15) Identify all prime implicants, essential prime implicants, and draw the final simplified circuit.	06	1	3
2b	Explain the core principle of the Non-Restoring Division Algorithm. Use the Non-Restoring Division Algorithm to perform the unsigned division of 18 by 5 Use 5-bit representation for the dividend and divisor. Show all intermediate steps clearly in a table.	06	1	3
2c	Explain why the Two's Complement method is the preferred way to represent and perform arithmetic on signed integers in digital systems, compared to Signed Magnitude or One's Complement. Using 8 bits, demonstrate how to represent the decimal numbers +45 and -45 in Two's Complement.	04	1	3
3a	Design a 1bit full adder, with the help of the truth table and expressions for Sum and Carry out. Using 1 bit full adder, design a 4bit Parallel Adder and calculate the time required to compute the 4bit sum.	08	2	3

3b	Describe the working principle of a JK master–slave flip-flop with suitable waveforms. Discuss the advantage and disadvantages over other types flip-flops.	08	2	3
	OR			
4a	Design a 4-bit CLA adder, with supporting expressions and circuits. Analyze the time required for computing addition of two 4 bit numbers.	08	2	3
4b	Discuss the application of SR Latch as switch debouncer, with the suitable waveforms and circuits.	08	2	3
5a	Design a binary synchronous counter using T flip-flops to count, in the sequence 000,100,111,010,011,000,...?	10	3	3
5b	Eliminate the redundant states for the state diagram given below and draw the minimized state diagram using row elimination or implication method. 	06	3	3
	OR			
6a	Design a clocked sequential circuit using JK flip-flops for the state diagram shown. Assign appropriate binary codes to the states (state assignment in straight binary sequence) 	10	3	3
6b	Design a 4 bit Ring counter and draw its timing diagram	06	3	3
7a	Discuss any eight generic Addressing modes supported by a processor, clearly illustrating the use of specific addressing mode with an example instruction and effective address calculation.	08	4	3
7b	ORIGIN 0000h MOV #1000,SP PUSH ITEM1 PUSH ITEM2 PUSH ITEM3 CALL SUB_1 ORIGIN 100h SUB_1:	08	4	4

PUSH FP

MOV SP, FP

PUSH R0

PUSH R1

ITEM1 DATAWORD 1000

ITEM2 DATAWORD 2000

ITEM3 DATAWORD 3000

Note: Make suitable Assumptions

OR

	What are condition codes ? Illustrate with examples the use and function of all the CPU flags. Indicate the value of all the flags, after the execution of ADD instruction.			

8a	MOV #15H,R0
	MOV #97H,R1
	ADD R0,R1,R7
		08	1	4

8b	Analyze the performance of Two and Three Address Processors, by writing ALP codes using two-address and three-address instruction formats to compute y= x3+5x+70, and comment on the efficiency of each approach	08	4	4
9a	Explain the structure and operation of SRAM and DRAM memories. Compare their performance and characteristics.	10	4	4
9b	Analyse the performance of three bus cpu datapath by writing the control sequence for the instruction ADD R1,R5,R7. Assume the instruction is stored in the memory.	06	4	3

OR

10a	Analyze the performance, by writing the control sequence for the execution of complete instruction, SUB R2,R1 ; R1 - R2 -> R1 using single bus cpu datapath, with the help of the neat diagram representing the architecture.	10	4	4
10b	Derive the Average Memory Access Time (AMAT) equation. Analyze the effect of hit ratio on system performance	06	4	3